Lab 11 – Intermediate Code Generation (Quadruple, Triple, Indirect Triple)

Aim

To generate quadruple, triple, and indirect triple representations for a given arithmetic expression.

Algorithm

1. Start the program.
2. Read an arithmetic expression.
3. Scan the expression from left to right.
4. Identify operators according to precedence (*, /, +, -).
5. For each operation:
    1. Generate a temporary variable.
    2. Store operator, operand1, operand2, and result.
6. Replace the evaluated subexpression with the temporary variable.
7. Continue until the complete expression is reduced.
8. Display:

    1. **Quadruple**
    2. **Triple**
    3. **Indirect Triple**

9. Stop.

C Program

```
#include <stdio.h>
#include <string.h>

int main() {
   char expr[50];
   char op[10], arg1[10], arg2[10], res[10];
   int count = 0;
   int i;

   printf("Enter expression (Example: a=b+c*d): ");
   scanf("%s", expr);

   for (i = 0; expr[i] != '\0'; i++) {
      if (expr[i] == '*' || expr[i] == '/' ||
         expr[i] == '+' || expr[i] == '-') {

         op[count] = expr[i];
         arg1[count] = expr[i - 1];
         arg2[count] = expr[i + 1];
         res[count] = 'A' + count;
```

```
            expr[i + 1] = res[count];
            count++;
        }
    }

    printf("\nQuadruple\n");
    printf("No\tOp\tArg1\tArg2\tResult\n");

    for (i = 0; i < count; i++) {
        printf("%d\t%c\t%c\t%c\t%c\n",
               i,
               op[i],
               arg1[i],
               arg2[i],
               res[i]);
    }

    printf("\nTriple\n");
    printf("No\tOp\tArg1\tArg2\n");

    for (i = 0; i < count; i++) {
        printf("%d\t%c\t%c\t%c\n",
               i,
               op[i],
               arg1[i],
               arg2[i]);
    }

    printf("\nIndirect Triple\n");
    printf("Pointer\tStatement\n");

    for (i = 0; i < count; i++) {
        printf("%d\t(%d)\n", i, i);
    }

    return 0;
}
```

Sample Input

```
a=b+c*d
```

Sample Output

Quadruple

No	Op	Arg1	Arg2	Result
0	*	c	d	A
1	+	b	A	B

Triple

No	Op	Arg1	Arg2
0	*	c	d
1	+	b	A

Indirect Triple

Pointer	Statement
0	(0)
1	(1)

Result

Thus the quadruple, triple, and indirect triple representations were generated successfully for the given expression.